

# VISIONARY FACTORY V9 — 16MP JPG ZIP EXPORT ENGINE PROFESSIONAL TECHNICAL MANIFEST V2.0

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This document describes the full VISIONARY FACTORY V9 16MP workflow in a premium, client-facing format. It is designed as a technical trust signal for marketplace, contributor, product advertising, and local batch image export workflows.

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## 💠 1. 16MP Upscale

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The engine performs an image upscale to approximately 16 megapixels while preserving the detected aspect-ratio category.

Supported final output targets are calculated dynamically from the selected format:

- 💠 16:9 → approximately 5333 x 3000 px
- 💠 21:9 → approximately 6110 x 2619 px
- 💠 9:16 → approximately 3000 x 5333 px
- 💠 1:1 → approximately 4000 x 4000 px

### 💠 Technical Insight:

The upscale uses LANCZOS interpolation, a high-quality resampling method designed to preserve smoother edges and cleaner transitions during enlargement. This is stronger than basic

resizing and helps the final image keep better structure after scaling.

#### 💡 Why it matters:

16MP is a practical export size for marketplace use, website assets, product visuals, client previews, and contributor-style image preparation where strong quality is needed without producing extremely heavy files.

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#### 💡 2. Support for 16:9, 21:9, 9:16, and 1:1

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The engine supports four key visual formats:

- 💡 16:9 horizontal cinematic format
- 💡 21:9 ultrawide premium cinematic format
- 💡 9:16 vertical social/ad format
- 💡 1:1 square marketplace/product format

#### 💡 Technical Insight:

Each format is detected from the input image ratio, then matched to a corresponding 16MP export target. This makes the output predictable and keeps the image aligned with the intended marketplace or advertising use case.

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#### 💡 3. Batch Aspect-Ratio Protection

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The engine blocks mixed aspect-ratio batches.

Allowed batch examples:

- ◆ only 16:9 images
- ◆ only 21:9 images
- ◆ only 9:16 images
- ◆ only 1:1 images

Blocked examples:

- ◆ 16:9 + 9:16 in the same batch
- ◆ 16:9 + 1:1 in the same batch
- ◆ 21:9 + 16:9 in the same batch
- ◆ 1:1 + 9:16 in the same batch

#### ◆ Technical Insight:

One batch must contain only one supported format category. This prevents inconsistent output sizes, wrong export dimensions, and accidental mismatched image packs.

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#### ◆ 4. Unsupported Aspect-Ratio Detection

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If the engine detects an unsupported image format, it stops processing and reports the exact image size and category.

#### ◆ Technical Insight:

Images that do not match 16:9, 21:9, 9:16, or 1:1 are marked as OTHER. This protects the workflow from accidental bad input and prevents unpredictable resizing.

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## 💠 5. EXIF Rotation Fix

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The engine automatically applies EXIF rotation correction before processing.

### 💠 Technical Insight:

Some images from phones and cameras store rotation inside metadata instead of being physically rotated. The engine uses EXIF-aware correction so the final export does not come out sideways or incorrectly oriented.

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## 💠 6. RGB Conversion

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All images are converted to RGB before JPG export.

### 💠 Technical Insight:

PNG, WEBP, and other formats may contain transparency or non-RGB color modes. JPG export requires a clean RGB image. This step ensures that every final output is compatible with standard JPG delivery.

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## ◆ 7. Mild Color Enhancement

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The engine applies subtle color enhancement.

### ◆ Technical Insight:

Color is gently boosted to make the image more visually attractive without creating an oversaturated or artificial look.

### ◆ Result:

The image gains stronger marketplace appeal while still remaining natural.

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## ◆ 8. Mild Contrast Enhancement

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The engine applies a mild contrast boost.

### ◆ Technical Insight:

Contrast is increased carefully to improve visual separation and overall image punch without crushing shadows or making highlights too harsh.

### ◆ Result:

The image appears more polished, clearer, and more commercial.

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## ◆ 9. Pixel-Level Highlight Rolloff

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The engine uses NumPy-based processing to soften overly bright pixel values.

💡 Technical Insight:

Bright values above the highlight threshold are rolled down using a controlled formula. Instead of allowing white areas to clip harshly, the script gently compresses overly bright values.

💡 Benefit:

This helps reduce blown-out highlights and protects visual detail in bright zones, reflections, glass, chrome, skies, white products, and polished surfaces.

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💡 10. PRODUCT AD POLISH

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The engine applies a final product advertising polish pass.

💡 Technical Insight:

A subtle additional contrast refinement is applied after highlight control. This gives the image a more finished commercial look and improves perceived clarity.

💡 Result:

The final image feels more prepared for product advertising, marketplace presentation, and client-facing delivery.

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## 💠 11. Anti-Plastic Subtle Film Grain

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The engine adds controlled subtle film grain using Gaussian noise.

### 💠 Technical Insight:

AI-generated or heavily processed images can appear too smooth, flat, or plastic. A small amount of organic noise adds micro-variation and helps reduce the artificial digital look.

### 💠 Benefit:

The image feels more natural, photographic, and less over-smoothed.

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## 💠 12. 30MB–40MB JPG Targeting

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The engine attempts to export each JPG inside the 30MB–40MB file-size range.

### 💠 Technical Insight:

The script searches through high JPG quality values and attempts to find a file size inside the target range. If the target cannot be reached naturally, it saves the best practical high-quality result.

### 💠 Important Note:

16MP JPG files do not always naturally reach 30MB–40MB. File size depends on image complexity, detail density, texture, grain,

and compressibility. Smooth or minimal images may stay smaller even at quality 100.

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### 13. JPG Export Only

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All supported input formats are exported as JPG.

Supported inputs:

-  JPG
-  JPEG
-  PNG
-  WEBP

Final output:

-  JPG only

#### Technical Insight:

JPG is widely compatible with marketplace platforms, websites, client previews, and advertising delivery systems. The engine exports every processed image into a standardized JPG format.

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### 14. High-Quality JPEG Settings

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The engine exports JPG files using high-quality settings.



Current export settings include:

- 💠 quality targeting
- 💠 subsampling=0
- 💠 300 DPI metadata
- 💠 in-memory JPEG encoding

💠 Technical Insight:

subsampling=0 helps preserve better color information in JPG export. 300 DPI metadata is included for professional presentation, although DPI metadata does not change the actual pixel resolution.

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💠 15. PASS / FAILED System

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Each processed file receives a success or failure status.

💠 PASS:

The file was processed and exported successfully.

💠 FAILED:

The file failed processing and the error is written into the report.

💠 Technical Insight:

This gives the batch workflow a safer professional structure. If one image fails, the report clearly shows which file had the issue and why.

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## 16. Dimension Validation

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The engine verifies that the exported JPG has the expected final dimensions.

### Technical Insight:

After saving the JPG, the engine reopens the file and checks its size. If the final dimensions do not match the intended 16MP target, the file is marked as failed.

### Benefit:

This protects against silent export mistakes.

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## 17. JPG Format Validation

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The engine verifies that the final output is truly a JPEG file.

### Technical Insight:

After export, the script reopens the saved file and checks the image format. If the output is not JPEG, the file is marked as failed.

### Benefit:

This ensures the final package contains proper JPG images only.

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## ◆ 18. TXT Report With Diamonds

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The engine generates a TXT report with diamond-style workflow documentation.

The report includes:

- ◆ batch format
- ◆ supported formats
- ◆ target dimensions
- ◆ approximate megapixels
- ◆ output format
- ◆ ZIP package name
- ◆ workflow steps
- ◆ input validation details
- ◆ original resolution
- ◆ final resolution
- ◆ JPG quality used
- ◆ final file size
- ◆ file-size target status
- ◆ PASS / FAILED status

### ◆ Technical Insight:

The TXT report works as a production manifest. It documents what happened to each file and makes the export package more professional.

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## 💠 19. ZIP Export Package

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The engine automatically creates a ZIP archive containing the final JPG files and the TXT report.

Final ZIP example:

💠 VISIONARY\_FACTORY\_V9\_16MP\_JPG\_EXPORT.zip

💠 Technical Insight:

A ZIP package is easier to deliver, archive, upload, sell, or hand off to clients. It keeps all output files and documentation together in one organized file.

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## 💠 20. Clean Folder Structure

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The engine uses a clean folder structure:

💠 1\_RAW\_INPUT

💠 2\_PROCESSING

💠 3\_READY\_FOR\_UPLOAD

💠 Technical Insight:

Input images are placed into 1\_RAW\_INPUT. Final JPG exports and the report are written into 3\_READY\_FOR\_UPLOAD. The processing folder is prepared for future workflow extensions.

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## 💠 21. Auto Cleanup Of Old Output

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Before each run, the engine clears old output folders and removes the previous ZIP package.

### 💠 Technical Insight:

This prevents old images from mixing with new exports. Every run starts clean, which makes the final ZIP accurate and easier to trust.

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## 💠 22. Multi-Core Batch Processing

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The engine uses `ProcessPoolExecutor` for batch processing.

### 💠 Technical Insight:

Multiple images can be processed more efficiently using multiple CPU processes. This is useful for larger batches and repeated production workflows.

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## 💠 23. Progress Bar

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The engine displays a live progress bar while processing.

### 💠 Technical Insight:

The tqdm progress bar makes the workflow easier to monitor. You can see how far the batch has progressed instead of waiting without feedback.

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## 💠 24. Marketplace-Safe Processing Philosophy

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The engine is designed around practical marketplace-safe enhancement.

Its goal is to improve:

- 💠 resolution
- 💠 color
- 💠 contrast
- 💠 highlight behavior
- 💠 commercial polish
- 💠 output consistency
- 💠 final delivery structure

while avoiding:

- 💠 mixed-format batch mistakes
- 💠 unsupported image ratios
- 💠 wrong output dimensions
- 💠 non-JPG delivery
- 💠 messy export folders
- 💠 undocumented processing

### 💎 Final Positioning:

This is not just a simple resize script. It is a structured 16MP JPG ZIP export engine designed for clean batch processing, contributor-style preparation, product advertising visuals, and professional local image delivery.

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### 💎 IMPORTANT PROFESSIONAL NOTE

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This V9 16MP version does not currently embed a real sRGB ICC profile into the final JPG export.

It converts images to RGB, but the script does not use ImageCms ICC embedding inside `img.save()`.

For a higher PRO version, the recommended upgrade is:

- 💎 add ImageCms import
- 💎 create real sRGB ICC bytes
- 💎 pass `icc_profile=icc_data` into `img.save()`
- 💎 report whether ICC is embedded

This would upgrade the export from simple RGB JPG output to Real sRGB ICC Marketplace Export.

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END OF MANIFEST V2.0
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